



Yakeen Plus (2025)

Short Practice Test - 05

DURATION : 60 Minutes

DATE : 13/10/2024

M. MARKS : 192

ANSWER KEY

PHYSICS

1. (4)
2. (1)
3. (4)
4. (4)
5. (3)
6. (4)
7. (2)
8. (2)
9. (2)
10. (2)
11. (3)
12. (4)

CHEMISTRY

13. (1)
14. (2)
15. (3)
16. (2)
17. (4)
18. (3)
19. (1)
20. (2)
21. (2)
22. (2)
23. (3)
24. (2)

BOTANY

25. (2)
26. (4)
27. (1)
28. (2)
29. (2)
30. (3)
31. (1)
32. (1)
33. (3)
34. (4)
35. (3)
36. (2)

ZOOLOGY

37. (4)
38. (4)
39. (2)
40. (1)
41. (4)
42. (1)
43. (3)
44. (1)
45. (2)
46. (2)
47. (4)
48. (1)

SECTION-(I) PHYSICS

1. (4)
Kirchhoff's law was recognized, it had been experimentally established that a good absorber is a good emitter, and a poor absorber is a poor emitter. Naturally, a good reflector must be a poor absorber.

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2. (1)
Heat loss by water = heat gain by ice
It is given that,
Mass of water, $m = 80$ gm
Specific heat of water, $s = 1$ cal/gm/°C
So, heat lost by water during cooling from 30°C to 0°C
 $\Delta H = ms\Delta T$
 $\Delta H = 80 \times 1 \times (30 - 0)$
 $\Delta H = 2400$ cal ... (1)
Heat gain by ice, $m_1 l = m_1 \times 80$ cal/gm ... (2)
Where, l is the latent heat of fusion m_1 is the mass of ice melted.
From equation (1) and (2)
 $2400 = 80 m_1$
 $m_1 = 30$ g
Hence, 30 g of ice is melted.

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3. (4)
For isobaric process $\Delta Q = nC_p\Delta T$ and $\Delta W = nR\Delta T$
So, $\frac{\Delta Q}{\Delta W} = \frac{C_p}{R} = \frac{C_p}{C_p - C_v} = \frac{1}{1 - \frac{C_v}{C_p}} = \frac{\gamma}{\gamma - 1}$

$$\left[\because \frac{C_p}{C_v} = \gamma \right]$$

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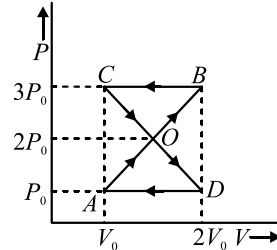
4. (4)
 $\frac{\Delta l}{l} \times 100\% = 3\%$
 $\alpha \Delta \theta = 0.03$
 $\alpha = \frac{0.03}{\Delta \theta}$
 $\therefore \beta = 2\alpha$
 $\beta = \frac{0.06}{\Delta \theta}$
 $\frac{\Delta A}{A} \times 100\% = \beta \Delta \theta \times 100\%$
 $= \frac{0.06}{\Delta \theta} \times \Delta \theta \times 100\% = 6\%$

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5. (3)
Internal energy does not depend on the path used to go from initial state to final state, but it depends on state.

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6. (4)
 $W = \text{area enclosed by } AODA + \text{by area enclosed } OBCO$



$$= \left[\frac{1}{2} \times (2V_0 - V_0) \times P_0 \right] + \left[-\frac{1}{2} (2V_0 - V_0) P_0 \right]$$

$$= 0 \left\{ \begin{array}{l} \text{AODA is a clockwise} \\ \text{while DBCO is anticlockwise} \end{array} \right.$$

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7. (2)
 $\therefore PV = RT$
 $\frac{V}{T} = \frac{R}{P} = \tan \theta \quad \left\{ \begin{array}{l} \because \text{Slope of } V-T \text{ graph} \\ m = \tan \theta = \frac{V}{T} \end{array} \right.$
i.e., $P = \frac{1}{\tan \theta}$
 $\theta_2 > \theta_1$ so $\tan \theta_2 > \tan \theta_1 \Rightarrow P_2 < P_1$, then $P_2 < P_1$
[New NCERT Class 11th Page No. 248]

8. (2)
For $\gamma = \frac{7}{5}$
 $C_v = \frac{R}{\gamma - 1} = \frac{R}{\frac{7}{5} - 1} = \frac{5R}{2}$
 $C_p = \frac{\gamma R}{\gamma - 1} = \frac{5R}{2} \times \frac{7}{5} = \frac{7R}{2}$
For $\gamma = \frac{5}{3}$
 $C_v = \frac{R}{\frac{5}{3} - 1} = \frac{3R}{2}$
 $C_p = \frac{3R}{2} \times \frac{5}{3} = \frac{5R}{2}$
 $\gamma_{\min} = \frac{2 \times \frac{7R}{2} + 1 \times \frac{5R}{2}}{2 \times \frac{5R}{2} + 1 \times \frac{3R}{2}} = \frac{7 + \frac{5}{2}}{5 + \frac{3}{2}} = \frac{19}{2} \times \frac{2}{13} = \frac{19}{13}$

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9. (2)

$$v_{rms} = \sqrt{\frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2}{5}} = \sqrt{11} \text{ km/s}$$

$$v_{mean} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3 \text{ km/s}$$

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10. (2)

$$\frac{P}{Q} = \frac{R}{S}$$

$$\frac{\frac{l}{k_4 A}}{\frac{l}{k_3 A}} = \frac{\frac{l}{k_2 A}}{\frac{l}{k_1 A}}$$

$$\frac{k_3}{k_4} = \frac{k_1}{k_2}$$

$k_2 k_3 = k_1 k_4$

[New NCERT Class 11th Page No. 216]

11. (3)

Cyclic process is anticlockwise then

Work done = – (Area of P - V graph)

$$W = -\pi R_1 R_2$$

$$= -\pi \left(\frac{3P_0 - P_0}{2} \right) \times \left(\frac{3V_0 - V_0}{2} \right)$$

$$= \frac{-22}{7} P_0 V_0$$

[New NCERT Class 11th Page No. 232]

12. (4)

0°C is 273 K so a body will radiate.

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SECTION-(II) CHEMISTRY

13. (1)

The hybridization state of central iodine atom in IF_5 , I_3^- and I_3^+ are sp^3d^2 , sp^3d , sp^3 respectively.

[New NCERT Class 11th Page No. 122]

14. (2)

Ionization energy increases along the period and decreases down the group.

Option (2) has $[\text{Ne}]3s^2 3p^3$, i.e., half-filled configuration, being more stable and thus, have high ionization energy

[New NCERT Class 11th Page No. 84]

15. (3)

By MOT

$$\text{N}_2 = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2$$

$$\text{N}_2^- = \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2$$

$$\pi^* 2p_x^1 \pi^* 2p_y^0$$

The electron adds to π^* orbital during change of N_2 to N_2^- .

[New NCERT Class 11th Page No. 126]

16. (2)

Ionic radii $\propto \frac{1}{Z_{\text{eff}}}$, Z_{eff} = effective nuclear charge

Ionic radii $\propto$ charge of anion

Ionic radii $\propto \frac{1}{\text{charge on cation}}$

Thus, the order of ionic radii is:

$$\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$$

[New NCERT Class 11th Page No. 87]

17. (4)

Polarizability $\propto$ Size of anion

Size of anion $\propto$ Polarisation $\propto$ Covalent character.

[New NCERT Class 11th Page No. 125]

18. (3)

- Cation has smaller size as compared to the size of parent atom generally.
- Electronegativity and ionisation enthalpy decreases on moving down the group of modern periodic table.
- In group 13, Al has more metallic nature than Ga due to shielding effect.
- Ne has greater ionization energy than Ar.

[New NCERT Class 11th Page No. 81]

19. (1)

SF ₆	⇒	Octahedral
NH ₃	⇒	Pyramidal
BF ₃	⇒	Trigonal planar
ClF ₃	⇒	T-shape

[New NCERT Class 11th Page No. 124]

20. (2)

The phenomenon in which the penultimate shell (n–1) electrons act as shield in between nucleus and valence shell electrons thereby reducing the effective nuclear charge is known as shielding effect.

[New NCERT Class 11th Page No. 81]

21. (2)

$$\text{Bond order} = \frac{1}{2}(N_b - N_A)$$

$$\text{B.O.} \Rightarrow \text{He}_2^+ = 0.5$$

$$\text{O}_2^- = 1.5$$

$$\text{C}_2^{2-} = 3$$

$$\text{Order will be} \Rightarrow \text{He}_2^+ < \text{O}_2^- < \text{C}_2^{2-}$$

[New NCERT Class 11th Page No. 129]

22. (2)

110 is named as Ununnilium.

[New NCERT Class 11th Page No. 100]

23. (3)

Generally electron affinity increases in a period and decreases in a group but due to smaller size and high electron density on fluorine atom, it experience high interelectronic repulsions. Hence, electron affinity is highest for chlorine. Its electronic configuration is

$$_{17}\text{Cl} = 1s^2, 2s^2, 2p^6, 3s^2, 3p^5$$

[New NCERT Class 11th Page No. 84]

24. (2)

Order of dipole moment is:

$$\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3 > \text{BF}_3$$

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SECTION-(III) BOTANY

25. (2)

- In the hypogynous flower the gynoecium occupies the highest position while the other parts are situated below it. The ovary in such flowers is said to be superior, e.g., mustard, china rose and brinjal.

[Old NCERT Class 11th Page No. 73]

26. (4)

In pea and bean flowers, there are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel); this type of aestivation is known as vexillary or papilionaceous.

[Old NCERT Class 11th Page No. 66]

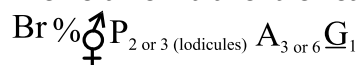
27. (1)

A bud is present in the axil of petiole in both simple and compound leaves.

[Old NCERT Class 11th Page No. 70]

28. (2)

The floral formula for the Poaceae family is



29. (2)

Sweet potato is not a modified storage tap root; instead, it is a modified adventitious root.

[Old NCERT Class 11th Page No. 67]

30. (3)

Pedicel	Stalk of flower
Bract	Reduced leaf
Petiole	Stalk of leaf
Pulvinus	Swollen leaf base

[Old NCERT Class 11th Page No. 70, 72]

31. (1)

Kelps, which are large brown algae, have cell walls primarily composed of cellulose, and they are also covered by algin.

[New NCERT Class 11th Page No. 27]

32. (1)

The spores in pteridophytes germinate to give rise to inconspicuous, small but multicellular, free-living, mostly photosynthetic thalloid gametophytes called prothallus.

[New NCERT Class 11th Page No. 30]

33. (3)

<i>Chlamydomonas</i>	Algae
<i>Cycas</i>	Gymnosperm
<i>Selaginella</i>	Pteridophyte
<i>Polytrichum</i>	Moss

[New NCERT Class 11th Page No. 26, 30, 31, 33]

34. (4)

The sex organs in bryophytes are multicellular.

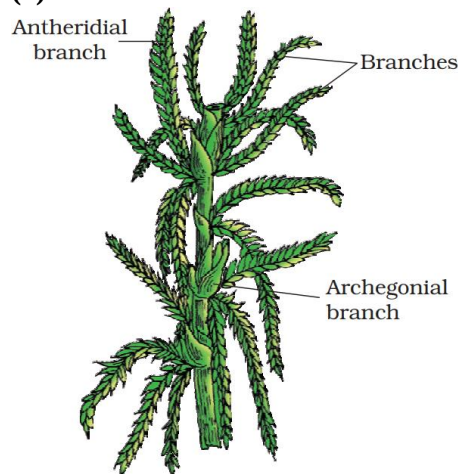
[New NCERT Class 11th Page No. 29]

35. (3)

Both statements are true and the reason provides a correct explanation for the assertion.

[New NCERT Class 11th Page No. 32]

36. (2)

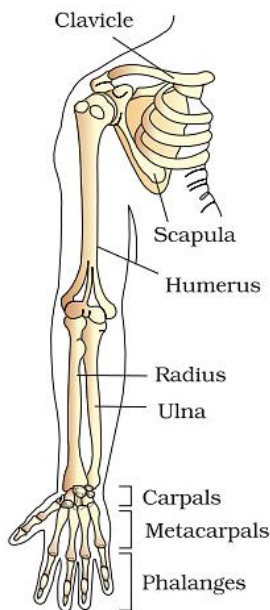


The above given diagram represents *Sphagnum*.

[New NCERT Class 11th Page No. 28]

SECTION-(IV) ZOOLOGY**37. (4)**

- Cardiac muscles are the muscles of the heart.
- Cardiac muscles are involuntary muscles that work continuously to pump blood throughout the body. They are branched and are striated in appearance.

[New NCERT Class 11th Page No. 118]**38. (4)****[New NCERT Class 11th Page No. 226]****39. (2)**

- Osteoporosis is an age-related disorder characterised by decreased bone mass and increased chances of fractures.
- Decreased levels of estrogen is a common cause.

[New NCERT Class 11th Page No. 223]**40. (1)**

The number of mitochondria are few in white muscle fibers.

[New NCERT Class 11th Page No. 223]**41. (4)**

- The repeated activation of the muscles leads to the anaerobic breakdown of glycogen.

- This takes place due to the unavailability of oxygen.
- This results in the formation of lactic acid, which accumulates and causes fatigue.

[New NCERT Class 11th Page No. 118]**42. (1)**

- Clavicle is the part of the appendicular skeleton.
- The skull, sternum, vertebral column and ribs make up the axial skeleton of the body.

[New NCERT Class 11th Page No. 224]**43. (3)**

Microfilament	Amoeboid movement
Visceral muscles	Non-striated
Axial skeleton	Skull
Tetany	Rapid spasm in muscle

[New NCERT Class 11th Page No. 218, 224, 227]**44. (1)**

During shortening of the muscle, i.e., contraction, the 'I' bands get reduced, whereas the 'A' bands retain the length.

[New NCERT Class 11th Page No. 222]**45. (2)**

- In the centre of each 'I' band is an elastic fibre called 'Z' line which bisects it.
- The globular head of myosin is an active ATPase enzyme and has binding sites for ATP.

[New NCERT Class 11th Page No. 220]**46. (2)**

Carpals (wrist bones— 8 in number) while metacarpals (palm bones – 5 in number)

[New NCERT Class 11th Page No. 218]

47. (4)

- If ATP is suddenly unavailable after the sarcomere has started to shorten (i.e., during muscle contraction), the cross-bridges formed between actin and myosin would not be able to detach because ATP is required for this detachment.

■ ■ ■

- Without ATP, the muscle remains contracted, leading to a state of rigidity.

[New NCERT Class 11th Page No. 222]

48. (1)

Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia.

[New NCERT Class 11th Page No. 219]



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